Chapter 5: Building Your Research

From Concepts to Testable Studies

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1 Overview

Before You Begin

This chapter bridges the gap between understanding experimental design concepts and actually planning your own research. You'll learn to turn abstract ideas into concrete, testable studies — the foundation of your Research Plan assessment.

This chapter covers:

- Operationalisation: Making variables measurable
- Writing hypotheses: One-tailed vs two-tailed choices
- Design decisions: Between-subjects vs within-subjects trade-offs
- Connecting design to analysis: Why your design determines your test
- Method section structure: What to include and why

By the end of this reading, you should feel prepared to start planning your Research Plan.

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2 Part 1: What is Operationalisation?

2.1 The Gap Between Ideas and Measurement

Research starts with ideas: *Does stress affect memory? Does eye contact build trust? Does hydration improve thinking?*

These are interesting questions, but they're not yet testable. To test them, we need to answer:

- What exactly do we mean by "stress" or "memory" or "trust"?
- How will we create different levels of our independent variable?
- How will we measure the dependent variable?

This is where operationalisation comes in.

Operationalisation: The process of turning an abstract concept into a concrete, measurable variable with a specific procedure for manipulation or measurement.

2.2 Why Operationalisation Matters

Good operationalisation enables:

1. **Replication**: Other researchers can repeat your study exactly
2. **Communication**: Readers understand precisely what you did
3. **Validity**: Your measure actually captures what you claim to measure
4. **Control**: You know exactly what differs between conditions

Poor operationalisation leads to:

1. **Ambiguity**: "We measured mood" — how? With what?
2. **Confounds**: Multiple things differ between conditions
3. **Irreplicability**: No one can reproduce your findings
4. **Invalid conclusions**: You measured something, but not what you think

2.3 Operationalising the Independent Variable

The IV is what you manipulate. You need to specify exactly how the conditions differ.

Vague: "Participants were either stressed or not stressed"

Operationalised: "Participants in the stress condition completed a timed mental arithmetic task (counting backwards from 2,043 by 17s) while being observed by an experimenter who maintained a neutral expression. Participants in the control condition sat quietly for the same duration (3 minutes) without any task."

Notice how the operationalised version tells you:

- What the stressor was (timed arithmetic)
- Specific parameters (starting number, subtraction amount)
- The social element (observed, neutral expression)
- The control condition (matched for time, no task)
- Duration (3 minutes)

2.4 Operationalising the Dependent Variable

The DV is what you measure. You need to specify the exact measurement procedure.

Vague: "We measured their memory"

Operationalised: "Memory was assessed using a free recall task. Participants were shown 20 words (10 concrete nouns, 10 abstract nouns) for 3 seconds each. After a 30-second filled delay (counting backwards from 50), participants wrote down as many words as they could remember in 2 minutes. The dependent variable was the total number of correctly recalled words (maximum = 20)."

This version specifies:

- The type of memory test (free recall)
- The stimuli (20 words, types specified)
- Presentation timing (3 seconds each)

- The retention interval (30 seconds, filled)
- The recall procedure (written, 2 minutes)
- The scoring method (total correct, max = 20)

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3 Part 2: The Three Research Plan Topics

Your Research Plan assessment requires you to design an experiment on one of three topics. Let's work through the operationalisation challenges for each.

3.1 Topic 1: Font Difficulty → Memory

The Research Question: Does reading text in a harder-to-read font improve memory retention?

Background: Diemand-Yauman et al. (2011) proposed that "desirable difficulties" — challenges that require more cognitive effort during learning — can enhance long-term retention. Reading disfluent fonts requires more effort, which might deepen processing.

3.1.a Operationalising "Font Difficulty"

You need two conditions that differ ONLY in font readability:

Condition	Font Specification
Fluent (Easy)	Arial, 16-point, black on white
Disfluent (Hard)	Comic Sans MS, 12-point, 75% grey on white

The key principle: Keep everything else constant (same words, same screen, same instructions) so that font is the only thing that differs.

Alternative operationalisations:

- Fluent vs Italic versions of the same font
- Standard vs reduced contrast
- Clear vs slightly blurred text

> ⚠ Potential Confound
>
> If your fonts differ in more than just readability (e.g., one is much smaller, or associated with different contexts), you can't be sure which difference caused any effect.

3.1.b Operationalising "Memory"

Common options:

Measure	Description	Pros	Cons
Free recall	Write down everything remembered	Sensitive to deeper processing	Floor/ceiling effects possible
Recognition	Multiple choice: "Was this word shown?"	Easy to score	May not detect subtle effects
Cued recall	Given category, recall specific items	Balanced difficulty	Requires careful stimulus design

Your choice should be justified based on prior research or theoretical considerations.

3.2 Topic 2: Hydration → Cognitive Performance

The Research Question: Does drinking water improve cognitive task performance?

Background: Edmonds & Burford (2009) found that children who drank water before cognitive tests performed better. Dehydration may impair attention and processing speed.

3.2.a Operationalising "Hydration"

Options for your IV manipulation:

Approach	Water Condition	Control Condition	Considerations
Simple manipulation	Drink 300ml water	No drink	Confound: act of drinking vs water
Placebo control	Drink 300ml water	Sip water (no swallowing)	Controls for drinking action
Standardised	Drink 250ml after 2-hour abstention	2-hour abstention, no drink	Controls prior hydration state

> ℹ Design Consideration
>
> Consider whether participants knowing they're in a "hydration" study might create expectation effects. If people believe water helps cognition, their performance might improve regardless of actual hydration.

3.2.b Operationalising "Cognitive Performance"

Options include:

- **Attention tasks**: Simple/choice reaction time, sustained attention (continuous performance task)

- **Working memory**: Digit span, n-back task
- **Processing speed**: Symbol-digit substitution, pattern comparison
- **Executive function**: Stroop task, Wisconsin Card Sorting Test

Your choice should match the theoretical mechanism. If hydration affects attention, use an attention task.

3.3 Topic 3: Eye Contact → Perceived Trustworthiness

The Research Question: Does eye contact duration affect how trustworthy a person is perceived to be?

Background: Argyle & Dean (1965) established that eye contact is a fundamental component of social interaction. Subsequent research has linked direct gaze to perceptions of competence, confidence, and trustworthiness.

3.3.a Operationalising "Eye Contact"

Options for manipulation:

Approach	High Eye Contact	Low Eye Contact	Considerations
Photographs	Eyes looking at camera	Eyes looking slightly away	Maximum control, less realistic
Videos	80% direct gaze	20% direct gaze	More realistic, harder to control
Duration	5 seconds of direct gaze	1 second of direct gaze	Tests duration specifically

Using photos or videos allows tight control — you can be sure the only difference is eye contact. Live interactions are more ecologically valid but introduce many confounds (voice, body language, natural variation).

3.3.b Operationalising "Trustworthiness"

Options:

Measure	Example	Pros	Cons
Single rating scale	"How trustworthy is this person?" (1-7)	Simple, direct	May not capture full concept
Multiple-item scale	Trust subscale from established measure	More reliable	Longer, may be redundant
Behavioural measure	"Would you lend this person £10?"	Face valid	Binary data less sensitive

Measure	Example	Pros	Cons
Economic game	How much would you invest with them?	Behavioural, continuous	Complex to administer

Multi-item scales are generally more reliable than single items. Consider using an established trustworthiness measure if one exists.

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4 Part 3: Writing Hypotheses

4.1 Review: Null and Alternative Hypotheses

Every statistical test compares your data against a null hypothesis.

Null Hypothesis (H_0): There is no effect — any observed difference is due to chance.

Alternative Hypothesis (H_1): There IS an effect — the observed difference is not due to chance.

For the font study:

- H_0: Font type has no effect on memory (μ_fluent = μ_disfluent)
- H_1: Font type does affect memory (μ_fluent ≠ μ_disfluent)

4.2 One-Tailed vs Two-Tailed Hypotheses

The alternative hypothesis can be:

Two-tailed (non-directional): Predicts there IS a difference, without specifying which direction.

> H_1: Memory scores will differ between fluent and disfluent font conditions.

One-tailed (directional): Predicts the specific direction of the difference.

> H_1: Memory scores will be higher in the disfluent font condition than the fluent font condition.

4.3 When to Use Each

Use Two-Tailed When…	Use One-Tailed When…
Effect could reasonably go either way	Strong theoretical basis for direction
Prior literature is mixed	Consistent prior findings support one direction
You're exploring a new question	Replicating established effect

Use Two-Tailed When…	Use One-Tailed When…
You want to detect unexpected effects	You're focused on confirming a specific prediction

4.4 Justifying Your Choice

For your Research Plan, you must justify why you chose one-tailed or two-tailed:

Example two-tailed justification:

> "A two-tailed hypothesis was used because prior research on disfluency effects has produced mixed results. While Diemand-Yauman et al. (2011) found memory benefits, subsequent replications have been inconsistent (Rummer et al., 2016). Given this uncertainty, we test for any difference rather than predicting a specific direction."

Example one-tailed justification:

> "A one-tailed hypothesis was used because the desirable difficulties framework (Bjork & Bjork, 2011) and the original disfluency research (Diemand-Yauman et al., 2011) both predict that harder-to-read fonts should enhance memory. We therefore predict that recall will be higher in the disfluent condition."

> ! Assessment Requirement
>
> Your Research Plan MUST include justification for your hypothesis direction choice. This demonstrates critical thinking about the research literature and your theoretical position.

4.5 Writing Complete Hypotheses

A complete hypothesis statement includes:

1. The IV and its levels
2. The DV and how it's measured
3. The predicted relationship

Complete two-tailed:

> "Participants who read a word list in a disfluent font (Comic Sans MS, 12pt, 75% grey) will show different free recall scores compared to participants who read the same word list in a fluent font (Arial, 16pt, black)."

Complete one-tailed:

"Participants who read a word list in a disfluent font (Comic Sans MS, 12pt, 75% grey) will recall significantly more words than participants who read the same word list in a fluent font (Arial, 16pt, black)."

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5 Part 4: Design Decisions — Between vs Within

5.1 The Fundamental Trade-off

You've chosen your topic, operationalised your variables, and written your hypotheses. Now you must decide how to assign participants to conditions.

Between-Subjects Design: Different participants in each condition. Each person experiences only ONE level of the IV.

Within-Subjects Design: Same participants in all conditions. Each person experiences EVERY level of the IV.

5.2 Between-Subjects Design

How it works:

- Randomly assign 30 participants to Condition A
- Randomly assign 30 different participants to Condition B
- Compare the mean scores of each group

Advantages:

1. **No order effects**: Participants can't be influenced by having done another condition first
2. **No practice effects**: Performance doesn't improve/decline across conditions
3. **No demand characteristics from comparison**: Participants don't try to be "consistent" or "different"
4. **Simpler procedure**: Each participant does one thing

Disadvantages:

1. **Individual differences add noise**: Smart people in one group inflate that mean
2. **Needs more participants**: Twice as many people for two conditions
3. **Pre-existing group differences**: Even with randomisation, groups might differ

5.3 Within-Subjects Design

How it works:

- All 30 participants complete Condition A
- Same 30 participants complete Condition B
- Compare each person's scores across conditions

Advantages:

1. **Controls individual differences**: Each person is their own control

2. **More statistical power**: Less "noise" makes real effects easier to detect
3. **Fewer participants needed**: Same people generate data for all conditions
4. **Can detect small effects**: Removing individual differences unmasks subtle differences

Disadvantages:

1. **Order effects**: First condition might affect second
2. **Practice effects**: Performance changes with repeated testing
3. **Fatigue effects**: Performance declines with longer sessions
4. **Demand characteristics**: Participants might guess what you expect

5.4 Design Decisions for Each Topic

5.4.a Font Difficulty → Memory

Design	Implementation	Key Concern
Between	Group A reads fluent, Group B reads disfluent	Individual differences in memory ability
Within	Everyone reads BOTH font versions	Memory for first list affects second

Typical choice: Between-subjects

Reasoning: If participants read the same words in both fonts, memory from the first reading will contaminate the second. If we use different word lists, we can't be sure the lists are equivalent in difficulty.

5.4.b Hydration → Cognitive Performance

Design	Implementation	Key Concern
Between	Group A drinks water, Group B doesn't	Individual differences in cognitive ability
Within	Same person tested hydrated and not hydrated	Requires two sessions or dehydration manipulation

Typical choice: Between-subjects (simpler) or within-subjects with two sessions (more powerful but logistically harder)

5.4.c Eye Contact → Trustworthiness

Design	Implementation	Key Concern
Between	Different participants see different gaze conditions	Individual differences in trust tendencies
Within	Same participants rate BOTH gaze conditions	Demand to rate them differently

Typical choice: Within-subjects with counterbalancing

Reasoning: If each participant rates multiple faces, we can have them see some faces with direct gaze and some with averted gaze. This controls for individual differences in how trusting people generally are.

5.5 Counterbalancing in Within-Subjects Designs

If you use a within-subjects design, you must address order effects through counterbalancing:

Counterbalancing: Varying the order of conditions across participants so that order effects cancel out.

- Half of participants: Condition A first, then Condition B
- Other half: Condition B first, then Condition A

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6 Part 5: Connecting Design to Statistical Test

6.1 The Critical Link

This section is essential for your Research Plan. You must demonstrate understanding of WHY your design choice leads to your test choice.

> ! Key Principle
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> The structure of your data (determined by your design) dictates which statistical test is appropriate.

6.2 Between-Subjects → Independent Samples t-test

The logic:

- You have two groups of participants
- Each participant provides one score
- The scores from Group A are independent of scores from Group B
- We compare the two group means

Why "independent"?: The scores in one group tell us nothing about the scores in the other group. Participant 1 in Group A has no connection to Participant 1 in Group B.

Example:

> "This study uses a between-subjects design with different participants randomly assigned to each font condition. The data therefore consist of two independent groups of recall scores. An independent samples t-test is the appropriate analysis because it compares the means of two unrelated groups."

6.3 Within-Subjects → Paired Samples t-test

The logic:

- You have one group of participants
- Each participant provides two scores (one per condition)
- The scores are paired — each person's Condition A score is linked to their Condition B score
- We compare the mean difference within each person

Why "paired"?: Participant 1′s score in Condition A is meaningfully connected to Participant 1′s score in Condition B — they come from the same person.

Example:

> "This study uses a within-subjects design with each participant rating faces in both gaze conditions. Because each participant provides paired scores (one trustworthiness rating for direct gaze, one for averted gaze), a paired samples t-test is appropriate. This test analyses whether the mean difference in ratings within individuals is significantly different from zero."

6.4 The Consequences of Using the Wrong Test

What happens if you use the wrong test?

Error	Consequence
Independent t-test on paired data	Loses statistical power (harder to detect real effects)
Paired t-test on independent data	Violates assumptions, produces invalid results

Both errors lead to incorrect conclusions about your research question.

6.5 Complete Justification Template

For your Research Plan, structure your justification like this:

> "This study employs a [between-subjects / within-subjects] design because [reason for design choice — reference advantages and how disadvantages are managed].
>
> Given this design, the data will consist of [description of data structure — independent groups OR paired scores].
>
> Therefore, [an independent samples / a paired samples] t-test is the appropriate statistical analysis. This test will compare [what is being compared] to determine whether [the research question]."

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7 Part 6: Method Section Structure

Your Research Plan requires a brief Method section. Here's what belongs in each subsection.

7.1 Design

State the design type and define your variables:

> "This study employed a between-subjects experimental design. The independent variable was font type with two levels: fluent (Arial, 16pt, black) and disfluent (Comic Sans MS, 12pt, 75% grey). The dependent variable was free recall score (number of words correctly recalled from a 20-word list, range 0-20)."

Must include:

- Design type (between-subjects, within-subjects)
- IV with levels fully operationalised
- DV with measurement details

7.2 Participants

Describe who participated and how they were assigned:

> "Sixty undergraduate psychology students will be recruited through the department participant pool. Participants will be randomly assigned to either the fluent font condition (n = 30) or the disfluent font condition (n = 30)."

Must include:

- Target sample size
- Recruitment method
- Assignment method (random, counterbalanced)

For within-subjects designs, specify counterbalancing:

> "Order of conditions will be counterbalanced, with half of participants seeing direct gaze faces first and half seeing averted gaze faces first."

7.3 Materials

Describe everything participants will see or use:

> "Stimuli consisted of 20 common concrete nouns selected from the MRC Psycholinguistic Database, matched for word frequency and imageability. In the fluent condition, words were displayed in 16-point Arial (black text on white background). In the disfluent condition, the same words were displayed in 12-point Comic Sans MS (75% grey on

white background). Presentation was controlled using PsychoPy. Participants recorded responses on a blank A4 sheet."

Must include:

- Stimulus details (what, how selected)
- Condition-specific differences
- Measurement tools/materials

7.4 Procedure

Describe what happens, step by step:

> "Participants will be tested individually in a quiet laboratory room. After providing informed consent, they will be seated at a computer and given standardised instructions to memorise the upcoming word list. Twenty words will be presented sequentially for 3 seconds each. After the final word, participants will count backwards from 50 for 30 seconds (filled delay), then write down as many words as they can remember for 2 minutes. The session will last approximately 10 minutes."

Must include:

- Setting
- Sequence of events
- Timing for each phase
- Key instructions given
- Total duration

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8 Part 7: Setting Up Your Results Table

Your Research Plan asks you to create a summary results table **without actual data**. This demonstrates you understand what you'll measure and how to organise it.

8.1 What the Table Should Show

Condition	n	M	SD
Fluent Font	30	—	—
Disfluent Font	30	—	—

Leave M and SD columns empty — you're showing the structure, not inventing numbers.

8.2 For Within-Subjects Designs

The table structure is similar, but you might also show the mean difference:

Condition	M	SD
Direct Gaze	—	—
Averted Gaze	—	—
Difference	—	—

Or present it as paired observations:

Participant	Direct Gaze	Averted Gaze	Difference
1-30	—	—	—
Mean	—	—	—

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9 Summary

Concept	Key Point
Operationalisation	Turning abstract concepts into concrete, measurable variables
IV operationalisation	Exactly how you create different conditions
DV operationalisation	Exactly how you measure the outcome
Two-tailed hypothesis	Predicts difference without direction — use when uncertain
One-tailed hypothesis	Predicts specific direction — use when theory supports it
Between-subjects	Different people in each condition — no order effects but individual differences
Within-subjects	Same people in all conditions — controls individual differences but order effects
Independent t-test	Compares means from different groups (between-subjects)
Paired t-test	Compares means from same people (within-subjects)
Design-test link	Your design structure determines which test is appropriate

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10 Check Your Understanding

Before the lecture, make sure you can answer these questions:

1. What is operationalisation and why does it matter for research?
2. For your chosen topic, how would you operationalise the IV and DV?
3. What's the difference between one-tailed and two-tailed hypotheses?
4. When would you choose each type of hypothesis?
5. What are the main advantages and disadvantages of between-subjects designs?
6. What are the main advantages and disadvantages of within-subjects designs?
7. Why does a between-subjects design require an independent t-test?
8. Why does a within-subjects design require a paired t-test?
9. What happens if you use the wrong statistical test?

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11 Further Reading

For additional depth:

- Diemand-Yauman, C., Oppenheimer, D. M., & Vaughan, E. B. (2011). Fortune favors the bold (and the italicized): Effects of disfluency on educational outcomes. *Cognition, 118*, 111-115.
- Edmonds, C. J., & Burford, D. (2009). Should children drink more water? The effects of drinking water on cognition in children. *Appetite, 52*, 776-779.
- Coolican, H. (2017). Chapter 2: Planning research. VLeBooks